

Exam: Time Series Visualization with Bokeh

This exam tests your ability to visualize time series data using the Bokeh library. You will be working with the **Daily Minimum Temperatures in Melbourne** dataset.

Dataset: `daily-minimum-temperatures-in-melbourne.csv`

The dataset contains daily minimum temperatures (in Celsius) recorded in Melbourne, Australia, from 1981 to 1990.



BokehJS 3.9.0 successfully loaded.

Dataset loaded: 3650 rows

Date range: 1981-01-01 to 1990-12-31

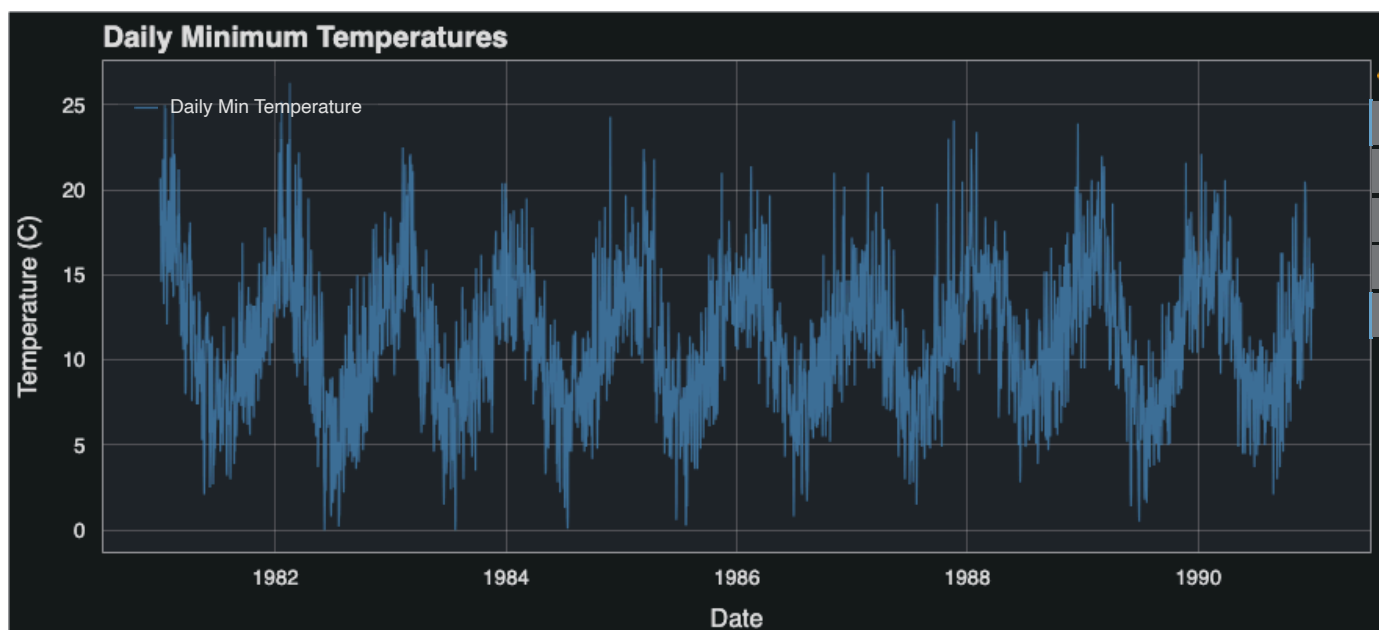
Temperature range: 0.0C to 26.3C

	Date	Temperature
0	1981-01-01	20.7
1	1981-01-02	17.9
2	1981-01-03	18.8
3	1981-01-04	14.6
4	1981-01-05	15.8

Question 1: Basic Time Series Line Plot

Create a basic line plot showing the daily minimum temperature over time.

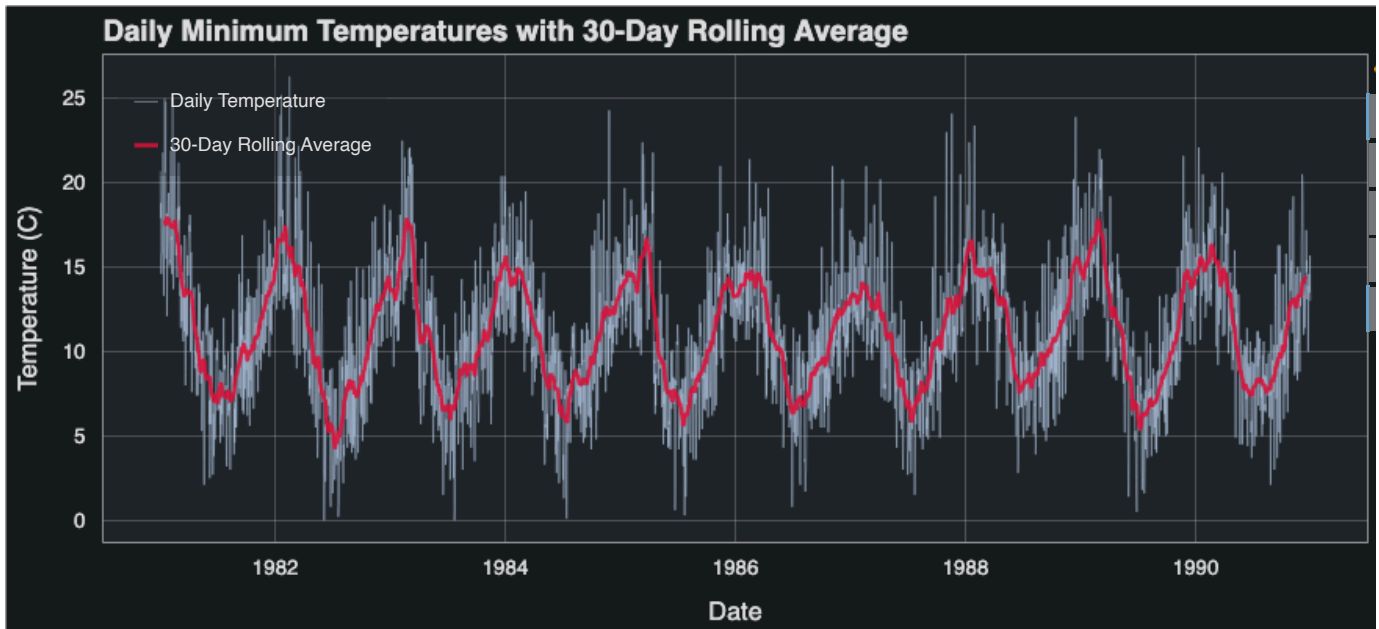
- Use the `Date` column on the x-axis and the `Temperature` column on the y-axis.
- Set the plot title to **"Daily Minimum Temperatures"**.
- Label the x-axis as `Date` and the y-axis as `Temperature (°C)`.
- Add tooltips to display the date and temperature when hovering over the line.
- Enable pan, wheel zoom, and reset tools.



Question 2: Rolling Average

Calculate the 30-day rolling average and plot it alongside the original temperature data.

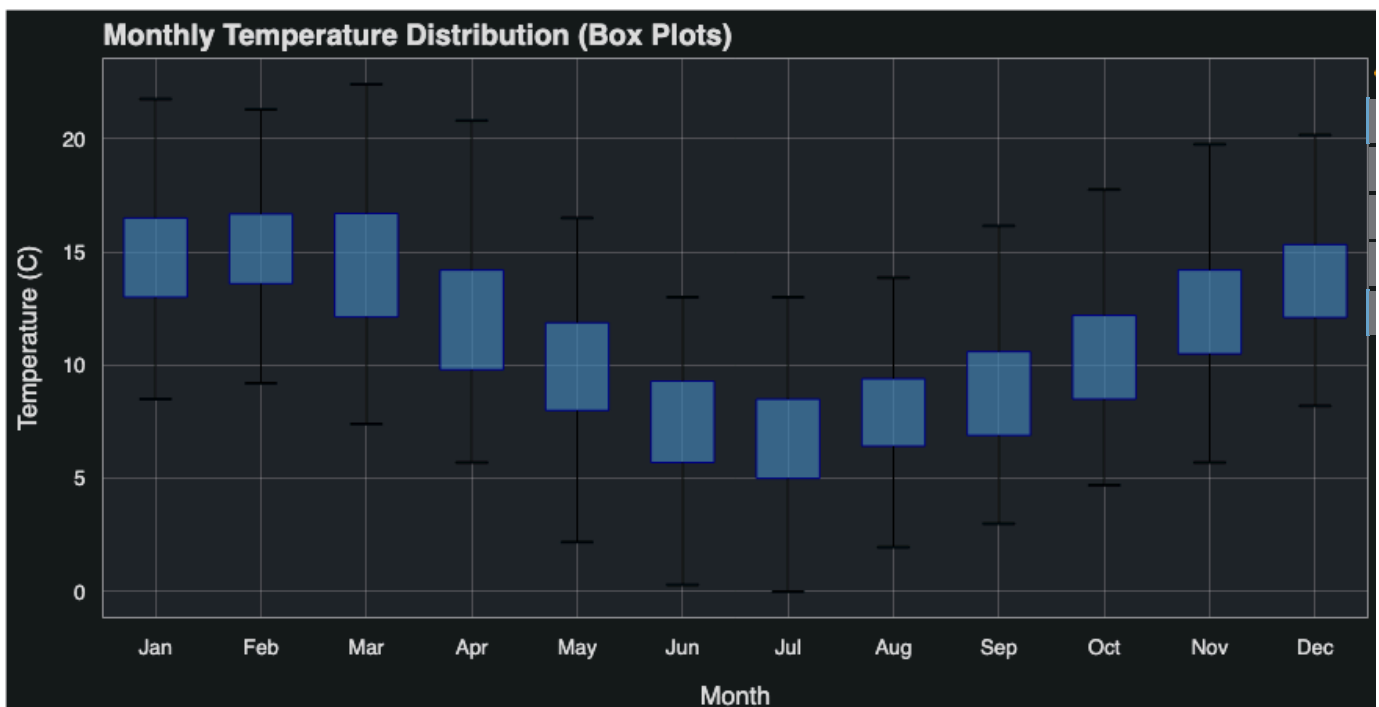
- Create a new column `Rolling_Avg` in the DataFrame.
- Plot both `Temperature` and `Rolling_Avg` on the same plot.
- Use different colors and line styles to distinguish the two.
- Add a legend and tooltips displaying date, temperature, and rolling average.



Question 3: Monthly Box Plots

Create box plots to visualize the distribution of temperatures for each month.

- Extract the month from `Date` and create a `Month` column.
- Use Bokeh's box plot elements (`vbar`, `segment`, `whiskers`).
- Label the x-axis with month names and y-axis with `Temperature (°C)`.
- Add tooltips displaying month and statistical values (min, max, median).

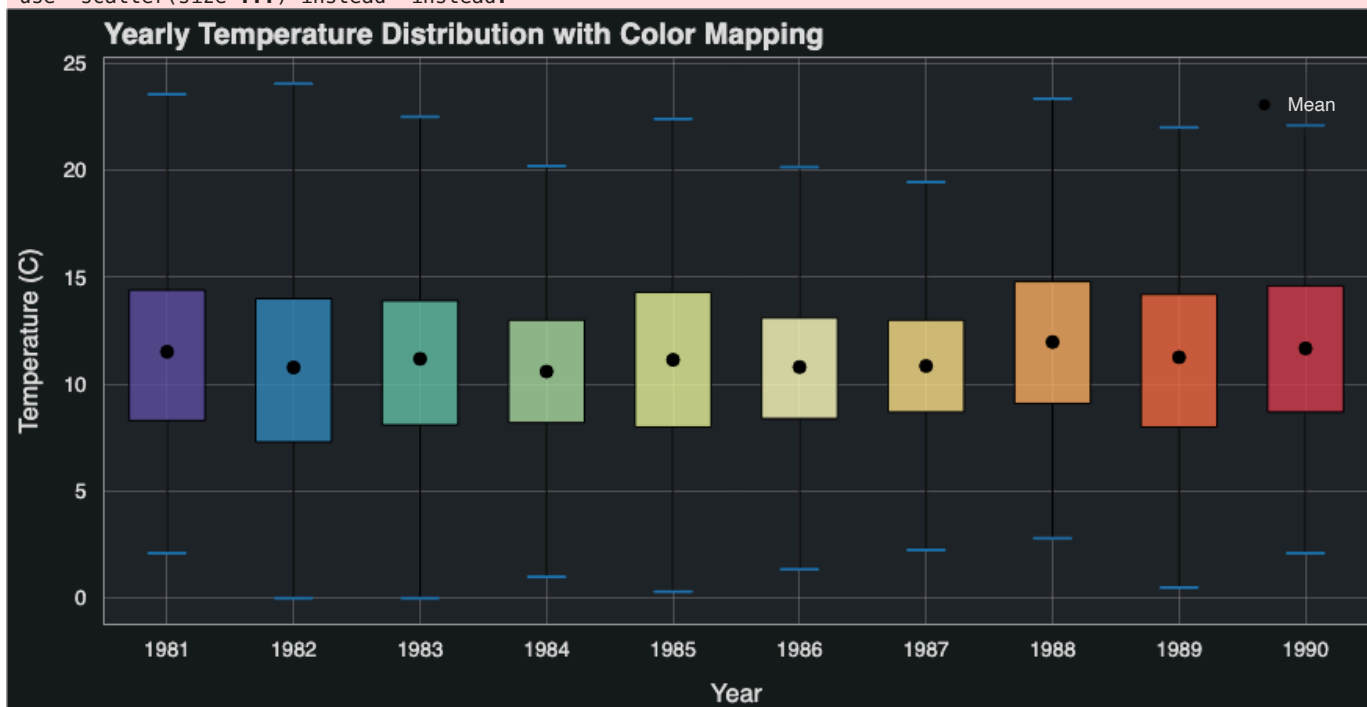


Question 4: Yearly Box Plots with Color Mapping

Create box plots to visualize the temperature distribution for each year.

- Extract the year from `Date` and create a `Year` column.
- Use `factor_cmap` to color boxes based on median temperature per year.
- Add tooltips with year and statistical values.
- Enable pan, wheel zoom, and reset tools.

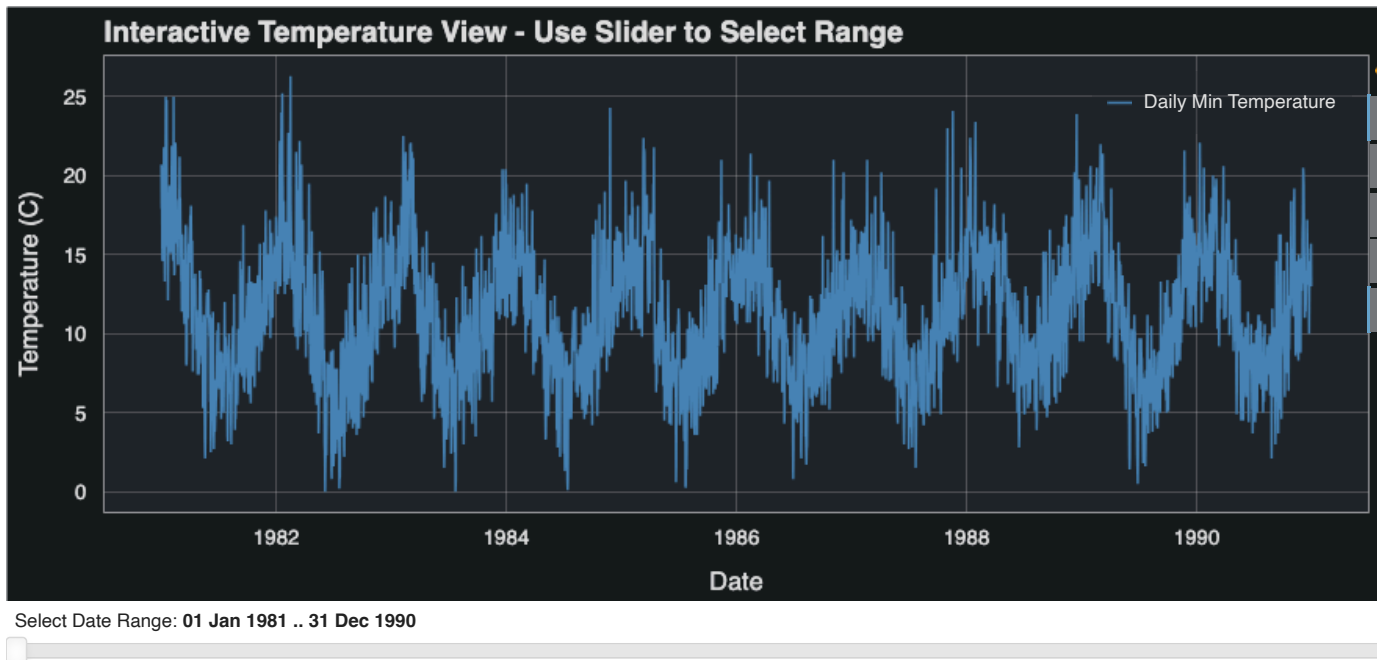
BokehDeprecationWarning: 'circle()' method with size value' was deprecated in Bokeh 3.4.0 and will be removed, use 'scatter(size=...)' instead' instead.



Question 5: Interactive Time Range Selection

Create an interactive line plot where the user can select a specific time range using a date range slider.

- Create a basic line plot of `Temperature` over `Date`.
- Implement a `DateRangeSlider` to filter the displayed date range.
- Update the plot dynamically based on the selected range.
- Add tooltips and enable pan, wheel zoom, reset tools.



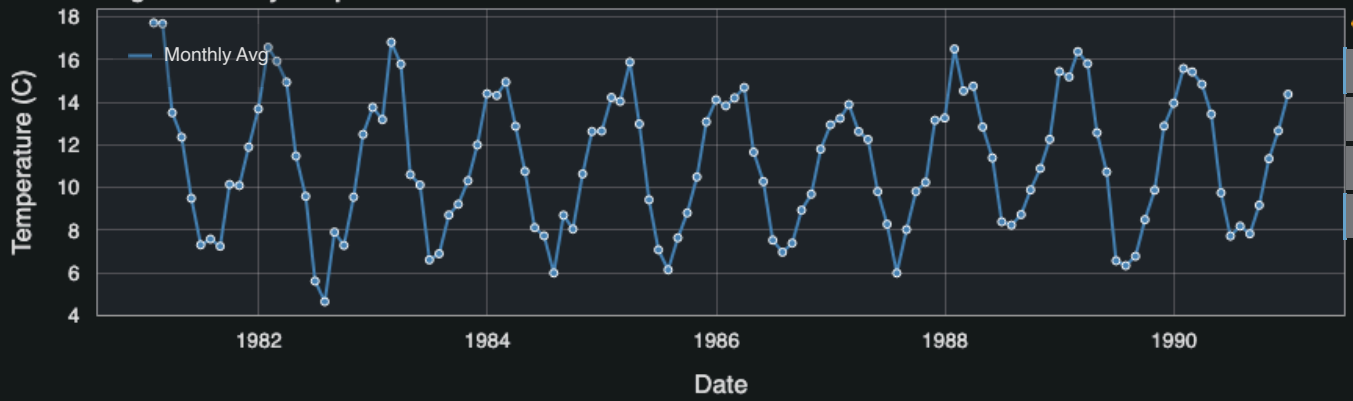
Question 6: Time Series Decomposition

Perform a simple time series decomposition to visualize trend and seasonality.

- Resample to monthly frequency and calculate monthly average temperature.
- Use a moving average to estimate the trend component.
- Calculate the seasonal component by subtracting the trend from the monthly data.
- Create three aligned Bokeh plots (original, trend, seasonal) sharing the same x-axis.
- Add tooltips to each plot and enable pan, wheel zoom, reset tools.

BokehDeprecationWarning: 'circle() method with size value' was deprecated in Bokeh 3.4.0 and will be removed, use 'scatter(size=...) instead' instead.

Original Monthly Temperature



Trend Component (12-Month Moving Average)



Seasonal Component (Monthly - Trend)

